10.2. Three principles for effective software

The following three principles all have keywords that make you ask, “Is our software effective?” The keywords are italicized in the following list of principles:

- *Working software* is the primary measure of progress.
- Our highest priority is to *satisfy the customer* through *early and continuous delivery* of valuable software.
- Continuous attention to *technical excellence and good design* enhances agility.

Working software is perhaps the most obvious of the software-related measures but many times the hardest to measure. Chapter 9 is dedicated to measuring how well your software is working.

Satisfying the customer is measured using techniques from chapter 6: using telemetry and business-specific metrics to measure how well your software does what it’s supposed to do.

Early and continuous delivery is covered in chapter 4 in our discussion of CI and deployment systems. This is enabled by build systems that output digestible reports and automate enough of your build and deployment cycle to get comprehensive data around build and deployment times. You can also measure this through your PTSs covered in chapter 3 by tracking lead time or development time.

You can argue that technical excellence and good design are measured by how fast you can iterate on your code or how maintainable it is, and how well it satisfies the consumer or how usable it is. Both of these are detailed in chapter 9.

10.2.1. Measuring effective software

Let’s start with a mind map to align the principles with the big question you’re trying to answer, as shown in figure 10.4.